Final Exam

Last Name: _____

First Name: _____

Student ID: _____

Instructions: This exam is due by 6pm on Friday, 20th March. Please typeset your solutions in latex, using the usual homework template. *Alternatively, you may also write out solutions by hand*. You will submit your solutions using Gradescope, in the same manner that you have submitted your homework assignments. This exam is open notes, which means that you may use all of the course material, as well as any textbooks or online resources, to complete it. This includes any content on our class Piazza page. However, **the work you submit must be your own**. In particular, discussing any question, in any way, with anyone other than your professor will be considered cheating. So, please do not contact your classmates during the examination window, and also please do not use any online Q&A forum like stackoverflow or Quora for assistance. If you have any questions about the exam, send your professor a private message on Piazza. For the computational questions, you may use MATLAB or Python, as long as you include your (well-commented) code as well as all required output. Obviously, due to the exceptional circumstances of this exam I cannot enforce these rules, so **I am trusting you to not cheat!**

Please do not write below this line.

Question	Points	Score
1	34	
2	36	
3	15	
4	25	
5	15	
6	35	
Total:	160	

1. Consider the following matrices and vectors:

$$A = \begin{bmatrix} 1 & 7 & 2 \\ 8 & 7 & 7 \\ 9 & 4 & 1 \\ 7 & 6 & 3 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 5 \\ 4 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

(a) (3 points) Compute $\| \mathbf{b} \|_1$

(b) (3 points) Compute $\lim_{p \to \infty} \| \mathbf{b} \|_p$. You do not need to show your work.

(c) (3 points) Compute $\|B\|_F$.

(d) In this part we will discuss the least squares problem:

$$\min_{\mathbf{x} \in \mathbb{R}^3} \| A\mathbf{x} - \mathbf{b} \|_2^2 \tag{1}$$

i. (10 points) Suppose that the QR factorization of A is:

$$Q = \begin{bmatrix} -0.0716 & 0.9056 & -0.3348 & -0.2505 \\ -0.5729 & 0.1753 & 0.7675 & -0.2282 \\ -0.6445 & -0.3618 & -0.5242 & -0.4230 \\ -0.5013 & 0.1354 & -0.1552 & 0.8404 \end{bmatrix} \quad R = \begin{bmatrix} -13.9642 & -10.0972 & -6.3018 \\ 0 & 6.9315 & 3.0829 \\ 0 & 0 & 3.7125 \\ 0 & 0 & 0 \end{bmatrix}$$

Use this information to solve (1). You may do this using a computer as long as you include well-commented code with your answer.

ii. (5 points) Name three other methods to solve, or approximately solve, (1) that we have discussed in class.

(e) Suppose that the rows of B represent Netflix users, while the columns represent movies on Netflix, so that $B_{ij} = \begin{cases} 1 & \text{if user } i \text{ has watched movie } j \\ 0 & \text{otherwise} \end{cases}$

i. (5 points) What is a non-negative matrix factorization of B? Is it unique?

ii. (5 points) How might we use a non-negative matrix factorization to better understand this data set?

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2. (a) (10 points) Write pseudocode for k-means.

   (b) The logistic regression classifier is $\sigma(\boldsymbol{\theta}^\top \mathbf{x} + b)$ where σ represents the sigmoid function and $\boldsymbol{\theta}, b$ are parameters to be learned.

      i. (5 points) Why do we use the sigmoid function here? Specifically, explain what purpose it is serving, and why it is a better choice than the step function $H(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$.

      ii. (5 points) What is the classification boundary (also known as the decision boundary)? Suppose that $\boldsymbol{\theta}, \mathbf{x} \in \mathbb{R}^2$, how would you find the equation of the classification boundary?

      iii. (6 points) How do we choose good values for the parameters $\boldsymbol{\theta}$ and b?

   (c) (5 points) Suppose that an oncologist[1] provides you with a liver cancer data set $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ with $\mathbf{x}^{(i)} \in \mathbb{R}^d$ and each entry $x_j^{(i)}$ recording a feature of the cancer, for example $x_1^{(i)}$ is the mass of the tumor, $x_2^{(i)}$ is the age of the patient and so on. The oncologist suspects there are two kinds of cancer present, but does not know how to distinguish them. Would you use k-means or logistic regression here? Justify your answer.

   (d) (5 points) Suppose that a cardiologist[2] provides you with a data set $\{(\mathbf{x}^1, y^1), \ldots, (\mathbf{x}^{(n)}, y^{(n)})\}$. Each $\mathbf{x} \in \mathbb{R}^d$ and each entry $x_j^{(i)}$ recording a feature of the patient, for example $x_1^{(i)}$ is the patient's weight, $x_2^{(i)}$ is the patient's blood pressure and so on and $y^{(i)} = 1$ if the patient has heart disease, and $y^{(i)} = 0$ otherwise. The cardiologist would like to devise a quick testing algorithm to decide whether or not someone has heart disease. Would you use k-means or logistic regression here? Justify your answer.

[1] cancer doctor
[2] heart doctor

3. (a) (10 points) Consider the standard form linear programming problem:

$$\min_{\mathbf{x}\in\mathbb{R}^4} \mathbf{c}^\top \mathbf{x} \text{ subject to: } A\,\mathbf{x} = \mathbf{b} \text{ and } \mathbf{x} \geq 0$$

where:

$$\mathbf{c} = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 0 \end{bmatrix}, A = \begin{bmatrix} -1 & 4 & -1 & 0 \\ 3 & 7 & 0 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} -1 \\ 5 \end{bmatrix} \tag{2}$$

Start with the basis $\{3, 4\}$ and perform one step of the simplex method. Clearly indicate the edge direction vectors $\boldsymbol{\delta}_j$, the unit costs $\mathbf{c}^\top \boldsymbol{\delta}_j$ the new basis and the new basic feasible solution $\mathbf{x}$. After this step, has the simplex algorithm converged? Justify your answer.

(b) (5 points) In class we discussed the two-phase simplex method. What is the purpose of the first phase?

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4. This question is on solving the problem:

$$\underset{\mathbf{x}\in\mathcal{X}}{\text{minimize}}\, f(\mathbf{x}) \tag{3}$$

For both parts, you may assume that $f(x)$ is continuously differentiable.

(a) For this part, assume that $\mathcal{X} = \mathbb{R}^d$, so the problem is unconstrained.

i. (5 points) Why is it convenient to assume that $f(\mathbf{x})$ is convex?

ii. (5 points) Suppose that $f(\mathbf{x})$ is convex, and that the dimension of the problem, d, is small. Would you use Newton's method or gradient descent to solve (3)? Why?

iii. (5 points) Now suppose again that $f(\mathbf{x})$ is convex, but that the dimension of the problem, d, is very large. Would you use Newton's method or gradient descent to solve (3)? Why?

(b) (10 points) For this part, assume that $\mathcal{X} = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 = 1\}$, *i.e.* the unit sphere. Write pseudocode for projected gradient descent. Include a formula for the projection operator as part of your pseudocode.

5. For *RNA*, the four nucleotide bases are U, C,G and A. Consider the following two strings of RNA:

$$X = UGACGA \quad \text{and} \quad Y = GUACGCA$$

(a) (10 points) Use dynamic programming to find the length of the longest common subsequence between X and Y. Include the table generated by the dynamic programming algorithm as part of your solution.

(b) (5 points) Now, use the traceback algorithm to find **all** longest common subsequences between X and Y.

You may find the code used to generate table 1 below useful:

a	b	c	d	e	f	g	h
i	j	k	l	m	n	o	p

Table 1: Always label your table!

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6. Consider the directed graph with adjacency matrix $A = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$.

   (a) (10 points) Sketch the graph. Number the vertices and draw in all directed edges, clearly showing their orientation. *I realize this may be a little tricky to do for a take home exam. You can either use the code used to create Figure 1, or you may sketch this by hand, take a picture/ scan it, and include it in your submission.*

   (b) (5 points) Compute $\tilde{P}$ using $\alpha = 0.1$. *Note that $\tilde{P}$ is as defined in homework 5.*

   (c) Below is pseudocode for the Power Method for determining the leading eigenvector, $\mathbf{v}_1$ of $\tilde{P}$. That is, $\mathbf{v}_1$ is the eigenvector corresponding to the largest eigenvalue of $\tilde{P}$.

Algorithm 1 The Power Method

Input: $\tilde{P}$, a random initial vector $\mathbf{v}^{(0)}$, number of iterations K
for $k = 1, \ldots, K$ **do**
  $\mathbf{v}^{(k)} \leftarrow \tilde{P}\,\mathbf{v}^{(k-1)}$
end for
Output: $\mathbf{v}^{(K)}$.

   i. (5 points) Explain, as we did in class, why the iterates $\mathbf{v}^{(k)}$ will converge to $\mathbf{v}_1$.

   ii. (5 points) How large should K be in order to guarantee $\| \mathbf{v}^{(K)} - \mathbf{v}_1 \|_2 \leq 0.0001$? Give your answer in terms of λ_2, the second-largest eigenvalue of $\tilde{P}$ *Hint: This will build on your work from part (i).*

   (d) (5 points) Using the Power Method compute $\mathbf{v}_1$. Use any $\mathbf{v}^{(0)}$ of your choice and $K = 1000$. *(If you choose to do this by hand, instead of using a computer, you may take $K = 3$)*

   (e) (5 points) Rank the vertices of this graph from most to least important.

Question 6 continues on the next page...

Question 6 continued...

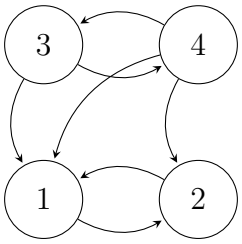


Figure 1: This figure is not related to this problem! I have just included it so that you can have access to the code used to create pictures of directed graphs. Make sure you include the two references to the `tikz` package in the preamble (the beginning of this tex file) in your file if you want to use this code.